MIL-M-46318A(MU)
16 FEBRUARY 1966
SUPERSEDING
MIL-M-46318 (Ord)
28 JUIY 1961

MILITARY SPECIFICATION

MOUNT, TELESCOPE: M101

1. SCOPE

1.1 This specification covers performance requirements, quality assurance provisions, and packaging data for a fire control instrument which mounts a direct fire telescope on self-propelled artillery vehicle (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and inspection of
MIL-P-14232	Parts, Equipment and Tools for Ordnance Materiel; Packaging of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions

DRAWINGS

U.S. Army Frankford Arsenal

F8242618	Mount, Telescope: M1O1

PACKAGING DATA SHEET

MIL-P-14232/P8242618	Packaging of Mount, Telescope: M101

FSC 1240

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(Copies of specifications, standards, drawings, and packaging data sheets required by suppliers in connection With specific Procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First article testing. - Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.2 Fabrication. - The mount shall be manufacture in accordance with Drawing F8242618 and drawing pertaining thereto.

3.3 General specification. - The contractor shall be responsible for adherence to the following requirements of MIL-F-13926.

(a) Dimensions and tolerances
(b) Inorganic protective surface finishes
(c) Part identification and marking
(d) Workmanship

3.4 Accuracy. - With the mount set up as shown on Drawing F8242616, the horizontal plane of the bottom surface and a line of sight through the geometric axis of the front and rear clamp assemblies shall form a angle of 2 degrees 15 minutes plus. or minus 1 mil and shall be parellel with the left side of the mount within limit.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirerments as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions. - The component and subassembly inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article testing. - When the contract specifies testing of preproduction or initial production samples, the following paragraphs shall apply as applicable.

4.2.1 Preproduction test. - Preproduction test samples shall be submitted in accordance with contract requirements. Unless otherwise specified in the contract, two preproduction samples shall be submitted to the procuring agency for approval of design and construction features.

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Preproduction samples Shall meet all inspection provisions of 4.3, 4.4 and 4.6 as well as any additional tests that may be specified in the contract.

4.2.2 Initial production test. - Initial production test samples shall be submitted in accordance with contract requirements. Unless otherwise specified in the contract, samples shall be the first two production units and shall have been manfactured using the same materials, processes and procedures as will be used in regular production. Initial production testing shall be in accordance with 4.3, 4.4 and 4.6 and shall be witnessed by the Government representative for all examinations and tests. The Government reserves the right to conduct any or all examinations or tests as may be deemed necessary.

4.2.3 First article disapproval.- Failure of any one of a first article sample submitted shall be cause for disapproval by the Government and subsequent acticn as required by the contract clauses applicable to first article testing.

4.3 Inspection provisions.

4.3.1 Submission of product. - Unless otherwise specified by the Contracting Officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests. - Components and subassemblies shall be inspected in accordance with the impaction provisions contained in MIL-F-13926. Examinations and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105, and the sampling plans specified in Tables I and II herein. Examinations and tests for Packaging, Packing, and Marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification- of defects shall constitute the minimum inspection to be performed by the supplier prior to Government acceptance or rejection by lot.

TABLE I - CLASSIFICATION OF DEFECTS

CLASS	REQUIREMENT	TEST PROCEDURE
CRITICAL: NONE		
MAJOR: AQL 1.5% Defective		
101. Accuracy	3.4	4.6.1
MINOR: NONE		

4.3.3 Acceptance and rejection. - Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

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* 4.4 Special sampling.

4.4.1 General. - One mount shall be selected at random as a special sampling from each 100 produced. The sample shall meet the requirements and tests in Table II.

TABLE II

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
301. Fabrication	3.2	Applicable drawings - visual
302. General specification	3.3	MIL-F-13926 - visual

4.5 Inspection equipment. - Except as otherwise provided for by the contractor, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection eqiuipment.- Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the epplicable Requirements specified in MIL-I-45607.

4.5.2 Contractor design. - The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures speclfied in "Examination" and "Test facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance iimit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration Or test equipment.

4.5.3 Zest equipment. - In conjunction with 4.5.2 the following special test equipment shall be utilized in the performance of the applicable tests as specified in 4.6.

4.5.3.1 Inspection equipment design requirements.- The special testing device used to simulate the telescope for testing the accuracy of the mount shall meet the following requirements:

(a) The device shall consist of a tube or round bar approximately fifteen inches in length. Two bearing rings conforming to the external dimensions shown on Drawing B8206346 shall be affixed to the tube or bar. The distance between ring centers shall be 11.875 inches plus or minus 0.005 inch. The rings shall be concentric to the tube or round bar within 0.0005 inch TIR (Totalindicator reading).

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(b) A finished measuring surface or location, located on the tube or round bar outside each bearing ring shall be used for measuring the anglular subtense of the 2 degrees 15 minutes angle formed by a line through the geometric centers of the clamps and the mounting surface of the mount.

4.6 Test methods and procedures.

4.6.1 Accuracy. The following tests shall be conducted with the special testing device specified in 4.5.3.1 inserted in the mount and sedued by means of the front and rear clamp assemblies.

4.6.1.1 Position the mount with its mounting surfaces in contact with a horizontal measuring surface them measure the angular subtense, of the angle formed by the mounting surface and the alinement of the fron and rear clamp assemblies. Measurements shall be made by using standard measuring equipment and the measuring surfaces SpeCified in 4.5.3.1(b). The angle formed shall not exceed the tolerance specified.

4.6.1.2 Position the mount on a horizontal measuring surface so that its mounting surface is in a perpendicular plane. Parallelism of a horizontal line through the geometric centers of the front and rear clamp assemblies to the bottom surface of the mount support shall not exceed the angular tolerance specified on Drawing F8242618, as measured by use of standrd measuring equipment.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking. - Packaging, packing, and marking shall be in accordance with MIL-P-14232 and Packaging Data Sheet Mil_p-14232/P8242618. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use. - The Telescope Mount: M101 is intended to support Telescope: M99C in a self-propelled artilery vehicle for use in direct fire operation.

6.2 Ordering daata. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Selection of applicable levels of preservation, packaging, and packing.
(c) Applicable packaging data sheet number (see 5.1).
(d) Applicable stock number.
(e) Order of document precedence.

Custodian:

Army - MU

Preparing activity:

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Project No. 1240-A464Y

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119 R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, activated in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the lowest cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION | CITY AND STATE

CONTRACT NO | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS (*Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity*)

SUBMITTED BY (*Printed or typed name and activity*) | DATE

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